Gavin Allison

Saskatoon, Saskatchewan

CONTACT

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SKILLS

Programming:
Python, TypeScript, NodeJS, SQL, Java

Web Development:
React, HTML, CSS, Tailwind CSS

Machine Learning:
Pytorch, OpenCV, Numpy, Pandas

EDUCATION

Master of Visual Computing

Simon Fraser University

2025 - Present

B.Sc. Honours Computer Science

University of Saskatchewan

2021 - 2024

WORK EXPERIENCE

Shoppers Drug Mart

Photo Technician *2019 - 2021*

- Maintained high-quality service
- Managed digital imaging workflows
- Controlled store inventory levels
- Collaborated efficiently within teams

SUMMARY

Computer Science Masters student specializing in full-stack development, computer vision, and machine learning. Experienced in building ML pipelines, researching object detection and developing applications.
View my work at https://gavin-allison.github.io/profile.

PROJECTS

Web App for Investing Backtesting

TypeScript, React, Tailwind CSS, PostgreSQL

- Designed an algorithm to accurately generate a stock portfolio from a series of hypothetical transactions
- Built a responsive and informative user interface capable of smoothly supporting thousands of items
- Integrated a PostgreSQL database to manage distinct groups of stocks and transactions for individual users
- Incorporated AI tools to provide users with enhanced control over adding and managing transactions

Object Detection Machine Learning Research Project

Python, PyTorch

- Conducted research alongside a computer vision expert to quantify the impact of annotation quality and inter-annotator agreement on object detection performance
- Proved that model performance deteriorates linearly as annotation quality decreases, highlighting the critical importance of high-quality data
- Determined that increased variability in annotator skill decreases a model's performance linearly
- Provided empirical evidence demonstrating the value of further research into inter-annotator agreement
- Presented a comprehensive research paper based on these findings to faculty and peers

Image Classification Segmentation Pipeline

Python, PyTorch

- Designed and implemented convolutional neural networks (U-Net and ResNet-18) in PyTorch for image segmentation and classification
- Developed a categorical parallel training method to drastically reduce training time with minimal accuracy loss
- Compiled and visualized results into interpretable charts to clearly communicate model performance
- Tuned hyperparameters and utilized pixel accuracy and mean Intersection over Union (mIoU) to evaluate the pipeline